Kali Linux and Applications Lab
SEMESTER – V

Course Code:	23CSE3038	**Course Credits:**	**02**
Teaching Hours / Week (L: T: P):	**4 hours /week**	**CA Marks:**	**50**
Total Number of Teaching Hours:		**END-SEM Marks:**	**50**

Course Pre-requisites: Basic understanding of computer network.

Course Description:

This course provides a comprehensive introduction to Kali Linux and cyber security, blending theoretical knowledge with practical hands-on experience. Participants will explore the fundamentals of Kali Linux, followed by advanced techniques for penetration testing and network security. The course also delves into various aspects of cyber security, from basic concepts to advanced strategies, enabling participants to identify and mitigate potential security threats effectively.

Course Objectives: (3 to 5)

By the end of this course, participants will:

1. Gain a comprehensive understanding of the Kali Linux operating system and its application in cyber security.
2. Acquire hands-on experience with tools for penetration testing and network security.
3. Learn both basic and advanced cyber security techniques and strategies.
4. Learn to detect and exploit vulnerabilities in networks and systems responsibly.
5. Utilize ethical hacking practices to enhance and implement robust security measures.

Course Outcomes: (Five) – Use Blooms Taxonomy

Upon successful completion of this course, participants will be able to:

1. Successfully install, configure, and navigate the Kali Linux operating system, utilizing its tools for various cyber security applications.
2. Conduct comprehensive penetration tests using advanced tools such as Nmap and Metasploit, identifying and exploiting system vulnerabilities.
3. Understand and apply wireless security protocols, WiFi cracking techniques, and network vulnerability assessments.
4. Execute and defend against various cyber security attacks, including SQL injection, phishing, DoS, ARP spoofing, and SSL stripping.
5. Apply ethical hacking principles to improve security measures, develop custom exploits, and prepare detailed reports on security assessments.

Learning Resources

Text Books:

1. Stephen Fletcher ,"Hacking with Kali Linux", A Beginner's Guide to Ethical Hacking.Ebook Available

2. Raphaël Hertzog, Jim O'Gorman, and Mati Aharoni "Kali Linux Revealed", Mastering the

Penetration Testing Distribution

Reference Books:

1. Robert W. Beggs,"Mastering Kali Linux for Advanced Penetration Testing" ISBN 978-1-78216-312-1,open source , Ebook Available

Sr No	Name of Experiment	CO
1	Introduction to Kali Linux interface, basic commands, file management command, System Information Commands, Networking commands, Process Management Commands, Compression Commands.	CO 1
2	Use a password strength tester to evaluate the strength of various passwords.	CO 1
3	Conduct a phishing attack simulation to understand social engineering techniques.	CO 4
4	Perform a network scanning and vulnerability detection using tools like Nmap.	CO 2
5	Use ARP spoofing techniques to intercept network traffic. Also, simulate man in the middle attack.	CO4
6	Use a hash cracking tool like John the Ripper to recover passwords.	CO4
7	Set up a Kali Linux environment for penetration testing.	CO2
8	Set up a simple web server and perform a basic SQL injection attack.	CO2
9	Set up a honeypot to detect and analyze potential intrusions.	CO5
10	Perform an SSL stripping attack to intercept and modify HTTPS traffic.	CO4
11	Conduct a denial of service (DoS) attack on a target system.	CO4
12	Perform a WiFi cracking exercise using tools like Aircrack-ng	CO3

MOOC/ Video Lectures available at:

https://www.youtube.com/watch?v=Qm7i7bCL8lQ

https://www.youtube.com/playlist?list=PLbJzrBainMsONrKSKQmy32Ol0W6xkqdSx

https://www.classcentral.com/course/youtube-the-complete-kali-linux-course-beginner-to-advanced-w-english-subtitles-65852

https://www.knowledgehut.com/blog/security/top-cyber-security-projects

ASSESSMENT AND EVALUATION PATTERN		
WEIGHTAGE	Continuous Assessment (CA)	End Semester Assessment (ESA)
CA - Assignments (PBL / Case Study / Presentation / Seminar / Group Discussions / Quiz / Test		
ESA - Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating		
Experiential Learning - Internship (Summer / Winter) / Industry Visit / Site Visit / Field Trips		

Project Based Learning – Mini (Minor) Project / Major Project															
Course Articulation Matrix (CO-PO Mapping) [Subject Title – Subject Code]															
COs	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	Relevant to National/ Global/Regional
CO-1	3	-	-	-	2	-	-	-	-	-		2	2		Global
CO-2	3	2	-	-	2	-	-	-	-	-		2	2		Global
CO-3	3	2	-	1	2	-	-	-	-	-		2	2		Global
CO-4	3	2	-	1	2	-	-	-	-	-		2	2		Global
CO-5	3	2	-	1	2	-	-	-	-	-	1	2	2		Global
3 – HIGH, **2** – MEDIUM, **1** – LOW															